EMERGENCY EVACUATION FLOW OPTIMIZER USING THE EDMONDS–KARP ALGORITHM

Project Report submitted for the partial fulfilment of grade for the course

Data Structures and Algorithms -3 (25CS2103E)

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ABSTRACT

This project presents an implementation of the Edmonds–Karp algorithm to analyze emergency evacuation flow during situations such as fires and floods using Breadth-First Search (BFS). The system models an evacuation network in which locations such as danger zones, corridors, junctions, and safe zones are represented as nodes, while evacuation routes are represented as directed edges with capacities indicating the maximum number of people that can pass through per minute. During execution, the application displays each augmenting path selected by BFS and illustrates how the residual graph changes after every iteration. These intermediate steps help users understand how evacuation flow is distributed through available routes and how capacity constraints affect the movement of people. The system also provides a visual representation of the evacuation network and reports the final maximum evacuation flow. By presenting the algorithm step by step in a real-world emergency scenario, the project demonstrates the practical application of graph algorithms to emergency management and makes the Edmonds–Karp algorithm easier to understand.

Signature of the Guide